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YOUR BRIDGE TO BRILLIANCE

UNIT 5

Impact of ml in BI Process

Business Intelligence

Evaluation of Classification Models

1. Evaluation of Classification Models

- Classification models are evaluated to measure their performance.
- Evaluation is based on predictions made by the model.
- Four important values are used:

TP **True Positive (TP)** – Actual positive and predicted positive.

TN **True Negative (TN)** – Actual negative and predicted negative.

FP **False Positive (FP)** – Actual negative but predicted positive.

FN **False Negative (FN)** – Actual positive but predicted negative.

		Actual Class: Positive	Actual Class: Negative
Predicted Class	Positive	**TP** (True Positive)	**FP** (False Positive)
	Negative	**FN** (False Negative)	**TN** (True Negative)

These values form the Confusion Matrix.

2. Accuracy

- Accuracy measures the overall correctness of a classification model.
- It shows the percentage of correct predictions out of all predictions made.
- Higher accuracy indicates better performance.

Formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Example:
Out of 100 predictions, if 90 are correct, Accuracy = 90%.

3. Precision

- Precision measures how many predicted positive cases are actually positive.
- It focuses on the quality of positive predictions.
- Important when False Positives are costly.

Formula:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Example:
If a model predicts 50 patients have a disease and only 40 actually have it, Precision = 40/50 = 80%.

Eg, predicting someone has disease when they don't.

4. Recall

- Recall measures how many actual positive cases are correctly identified.
- It focuses on finding all positive instances.
- Important when False Negatives are costly.

Formula:

$$\text{Recall} = \frac{TP}{TP + FN}$$

Example:
If 100 people have a disease and the model identifies 90, Recall = 90%.

Eg, missing a disease.

5. F1-Score

- F1-Score combines Precision and Recall into a single metric.
- It is the harmonic mean of Precision and Recall.
- Useful when both Precision and Recall are important.

Formula:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Example:
Precision = 80%
Recall = 90%
F1-Score ≈ 84.7%

6. Importance of Evaluation Metrics

- **Accuracy** → Overall correctness.
- **Precision** → Correctness of positive predictions.
- **Recall** → Ability to find actual positive cases.
- **F1-Score** → Balance between Precision and Recall.

These metrics help compare and improve classification models.

Bayes' Theorem

- Bayes' Theorem is a fundamental concept in probability and statistics.
- It is used to calculate the probability of an event based on prior knowledge.
- It helps update our belief when new information becomes available.
- It is useful for reasoning and decision-making under uncertainty.

Need for Bayes' Theorem

- Helps make predictions using available evidence.
- Updates probabilities when new data is received.
- Works well with incomplete or uncertain information.
- Calculates the probability of a cause given an observed effect.

Example:

- Determining the probability of having a disease after receiving a positive te

Bayes' Theorem

Formula of Bayes' Theorem

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

Where:

- P(A|B) = Probability of event A occurring given B has occurred.
- P(B|A) = Probability of event B occurring given A has occurred.
- P(A) = Prior Probability of A.
- P(B) = Probability of B (Evidence).

Applications of Bayes' Theorem

- Medical Diagnosis
- Spam Email Filtering

Bayes' Theorem

- Machine Learning
- Risk Analysis
- Weather Forecasting
- Fraud Detection

Example:

- Email services use Bayes' Theorem to classify emails as spam or non-spam.

Medical Diagnosis Example

Given:

- Disease affects 4 out of 1000 people.
- Test detects disease correctly 99% of the time.
- False positive rate = 5%.

an simply looking at the test accuracy.

Bayes' Theorem

- "What is the probability that a person actually has the disease if the test result is positive?"
- This gives a more realistic estimate than simply looking at the test accuracy.

Advantages of Bayes' Theorem

- Handles uncertainty effectively.
- Uses prior knowledge and new evidence together.
- Improves decision-making.
- Provides more accurate probability estimates.
- Widely used in AI, Data Science, and Statistics.

Naive Bayes Classifier

- Naive Bayes is a Machine Learning classification algorithm.
- It is based on Bayes' Theorem.
- It assumes that all features are independent of each other.
- It predicts the class with the highest probability.
- It is widely used for classification tasks.

Working of Naive Bayes Classifier

- Takes input features from the dataset.
- Calculates the probability of each class using Bayes' Theorem.
- Assumes features are independent.
- Compares probabilities of all classes.
- Assigns the class with the highest probability as the prediction.

Naive Bayes Classifier

$$P(C|X) = \frac{P(X|C) \times P(C)}{P(X)}$$

Where:

- P(C|X) = Probability of class C given features X
- P(X|C) = Probability of features X given class C
- P(C) = Prior probability of class C
- P(X) = Probability of features X

Applications of Naive Bayes

- Spam Email Detection
- Sentiment Analysis
- Text Classification

Naive Bayes Classifier

- News Categorization
- Recommendation Systems
- Medical Diagnosis

Example:

- Classifying an email as Spam or Not Spam.

Advantages of Naive Bayes

1) Simple and Fast

- Easy to implement.
- Requires less computational power.
- Trains quickly even on large datasets.

2) Performs Well with Less Data

- Gives good results even with small training datasets.

Naive Bayes Classifier

- Suitable when limited data is available.

3) Robust to Irrelevant Features

- Can handle extra or unnecessary features.
- Accuracy is not greatly affected.

4) Good for Real-Time Prediction

- Fast prediction speed.
- Suitable for real-time applications such as spam filtering.

Disadvantages of Naive Bayes

1) Not Suitable for Complex Relationships

- Assumes features are independent.
- Cannot capture complex or non-linear relationships between features.

2) Continuous Data Needs Assumptions

- Works naturally with categorical data.
- For continuous data, a distribution (such as Gaussian) must be assumed.
- The assumption may not always be correct.

Logistic Regression

- Logistic Regression is a popular Machine Learning algorithm used for classification problems.
- It is mainly used when the target variable is binary (Yes/No, Pass/Fail, True/False).
- Unlike Linear Regression, it predicts probabilities instead of continuous values.
- It estimates the likelihood that an input belongs to a particular class.

Working of Logistic Regression

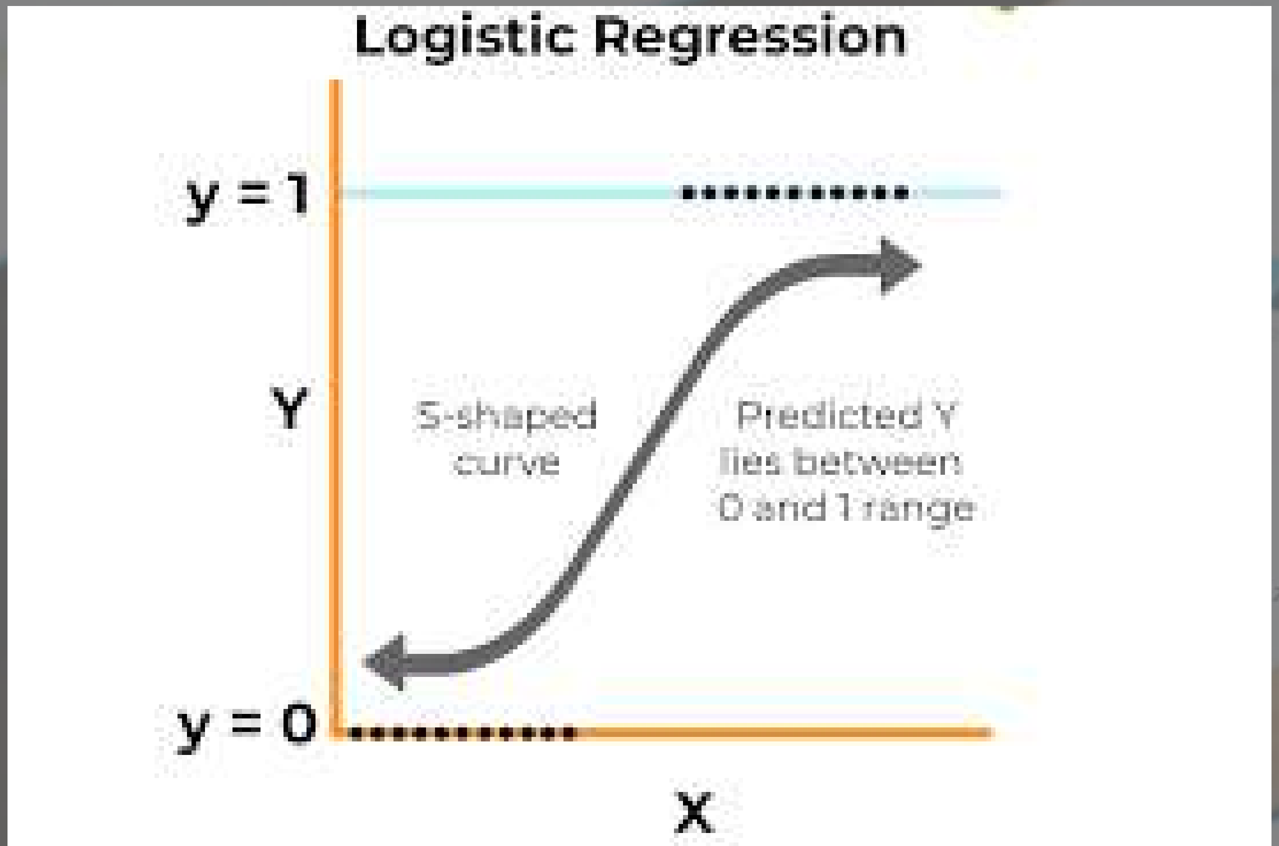
- Takes input features from the dataset.
- Calculates a weighted sum of the inputs.
- Applies the Sigmoid (Logistic) Function.
- Produces an output between 0 and 1.
- The output is interpreted as probability.
- Based on a threshold (usually 0.5), the final class label is assigned.

Logistic Regression

- Sigmoid Function

$$y = \frac{1}{1 + e^{-x}}.$$

- Output always lies between 0 and 1.
- Converts any real value into a probability.



Example of Logistic Regression

Problem: Predict whether a student will Pass or Fail based on study hours.

- Input: Hours Studied
- Predicted Probability = 0.85
- Threshold = 0.5

Since:

- 0.85 > 0.5

Prediction: Student is likely to Pass.

This makes Logistic Regression useful for decision-making based on probabilities.

Logistic Regression

Training of Logistic Regression

- The model learns from historical data.
- It adjusts the weights of input features.
- The goal is to minimize prediction errors.
- Training is performed using Maximum Likelihood Estimation (MLE).
- Better weights lead to more accurate predictions.

Types of Logistic Regression

1) Binary Logistic Regression

- Used when there are only two possible classes.
- Most commonly used type.

Examples:

- Pass / Fail
- Yes / No
- Spam / Not Spam

2) Multinomial Logistic Regression

- Used when the output has more than two categories.

Logistic Regression

- Categories are not ordered.

Example:

- Predicting Fruit Type:
    - Apple
    - Banana
    - Orange

based on features such as color and weight.

Ordinal Logistic Regression

- Used when the output has more than two categories.
- Categories have a natural order or ranking.
- Example: Customer Satisfaction
- Low
- Medium
- High

Clustering

- Clustering is an Unsupervised Learning technique in Machine Learning.
- It is used to group similar data points together.
- Each group formed is called a Cluster.
- Data points within the same cluster are more similar to each other than to points in other clusters.
- Clustering helps discover hidden patterns in data.

K-Means Clustering

- K-Means is one of the simplest and most widely used clustering algorithms.
- The letter K represents the number of clusters to be formed.
- It divides a dataset into K groups based on similarity.
- The algorithm tries to minimize the distance between data points and the center of their assigned cluster.
- The center of a cluster is called a Centroid.

Clustering

Example of K-Means Clustering

- Customer Segmentation
- A shop wants to group customers based on:
- Annual Income
- Spending Habits
- Using K-Means, customers can be divided into groups such as:
- Low Income – Low Spending
- Low Income – High Spending
- High Income – Low Spending
- High Income – High Spending
- This helps businesses understand customer behavior and make better marketing decisions.

Clustering

Step 1 & Step 2 of K-Means

Step 1: Choose the Number of Clusters (K)

- Decide how many clusters are needed.
- Example: K = 3 means three clusters will be created.

Step 2: Initialize Centroids

- Randomly select K data points from the dataset.
- These points act as the initial centroids (cluster centers).

Step 3 & Step 4 of K-Means

Step 3: Assign Data Points to Nearest Centroid

- Calculate the distance between each data point and every centroid.
- Assign the point to the nearest centroid.

Step 4: Recalculate Centroids

- After assigning all points, calculate the mean of points in each cluster.
- The mean becomes the new centroid.

Step 5

- Step 5: Repeat the Process
- Repeat Steps 3 and 4.
- Continue until centroids stop changing significantly.

Frequent Item Set

- A Frequent Item Set is a group of items that frequently appear together in a dataset.
- It is mainly used in Market Basket Analysis.
- It helps businesses identify products that customers often purchase together.
- Frequent itemsets reveal buying patterns and customer behavior.

Example:

- If many customers buy Bread and Butter together,
- Then {Bread, Butter} is a Frequent Item Set.

Applications of Frequent Item Sets

- Market Basket Analysis
- Product Recommendation Systems
- Cross-Selling Strategies
- Customer Behavior Analysis
- Inventory Management

Common Algorithms Used

- Apriori Algorithm
- FP-Growth Algorithm

These algorithms help discover frequent itemsets from large databases.

Frequent Item Set

- Minimum Support Count
- Minimum Support Count is a threshold used in Frequent Itemset Mining.
- It specifies the minimum number of times an itemset must appear in the dataset to be considered frequent.
- Itemsets with support below the threshold are ignored.
- Formula

$$\textit{Support Count}(\textit{Itemset}) = \text{Number of transactions containing the itemset}$$

Example of Minimum Support Count

Suppose:

- Minimum Support Count = 3

Itemset	Support Count
Bread, Butter	5
Milk, Bread	4
Tea, Sugar	2

Frequent Item Set

- Result
- Bread, Butter → Frequent ✔
- Milk, Bread → Frequent ✔
- Tea, Sugar → Not Frequent ✘
- Because only itemsets appearing 3 or more times are selected.

Hierarchical Clustering

- Hierarchical Clustering is a clustering technique that creates clusters in a tree-like structure.
- The tree structure is called a Dendrogram.
- It shows the relationship between clusters at different levels.
- Unlike K-Means, the number of clusters does not need to be specified initially.

Types of Hierarchical Clustering

Types of Hierarchical Clustering

1) Agglomerative Clustering (Bottom-Up)

- Starts with each data point as an individual cluster.
- Gradually merges the closest clusters.
- Continues until all points belong to a single cluster.

2) Divisive Clustering (Top-Down)

- Starts with all data points in one large cluster.
- Repeatedly splits the cluster into smaller clusters.
- Continues until each data point forms its own cluster.

Apriori Algorithm

- Apriori is a classic data mining algorithm.
- It is used to find Frequent Itemsets and generate Association Rules.
- It is widely used in Market Basket Analysis.
- The goal is to discover items that are frequently purchased together.
- It helps businesses understand customer buying patterns.

Apriori Principle

"If an itemset is frequent, then all of its subsets must also be frequent."

Working of Apriori Algorithm

Step 1: Find Frequent 1-Itemsets

- Scan the dataset.
- Count the occurrence of each item.
- Keep only items that satisfy the minimum support threshold.

Step 2: Generate Candidate 2-Itemsets

- Combine frequent 1-itemsets.
- Calculate their support count.
- Remove itemsets below the minimum support threshold.

Apriori Algorithm

Continue Itemset Generation

Step 3: Generate Larger Itemsets

- Combine frequent itemsets to form larger itemsets.
- Check their support count.
- Repeat until no more frequent itemsets can be generated.

Step 4: Pruning

- Remove infrequent itemsets.
- If an itemset is infrequent, its larger combinations are also removed.
- This reduces computation and improves efficiency.

Apriori Algorithm

Association Rule Generation

- After finding frequent itemsets, Apriori generates association rules.
- These rules show relationships between products.

Example Rule

If Milk is bought → Bread is also bought

This helps businesses understand customer purchasing

Slide 5: Example of Apriori Algorithm

Transactions

Transaction ID	Items
T1	Milk, Bread
T2	Milk, Bread, Butter
T3	Bread, Butter
T4	Milk, Butter

Support Count

Item	Count
Milk	3
Bread	3
Butter	3

Frequent Itemsets

- {Milk, Bread}
- {Bread, Butter}
- {Milk, Butter}

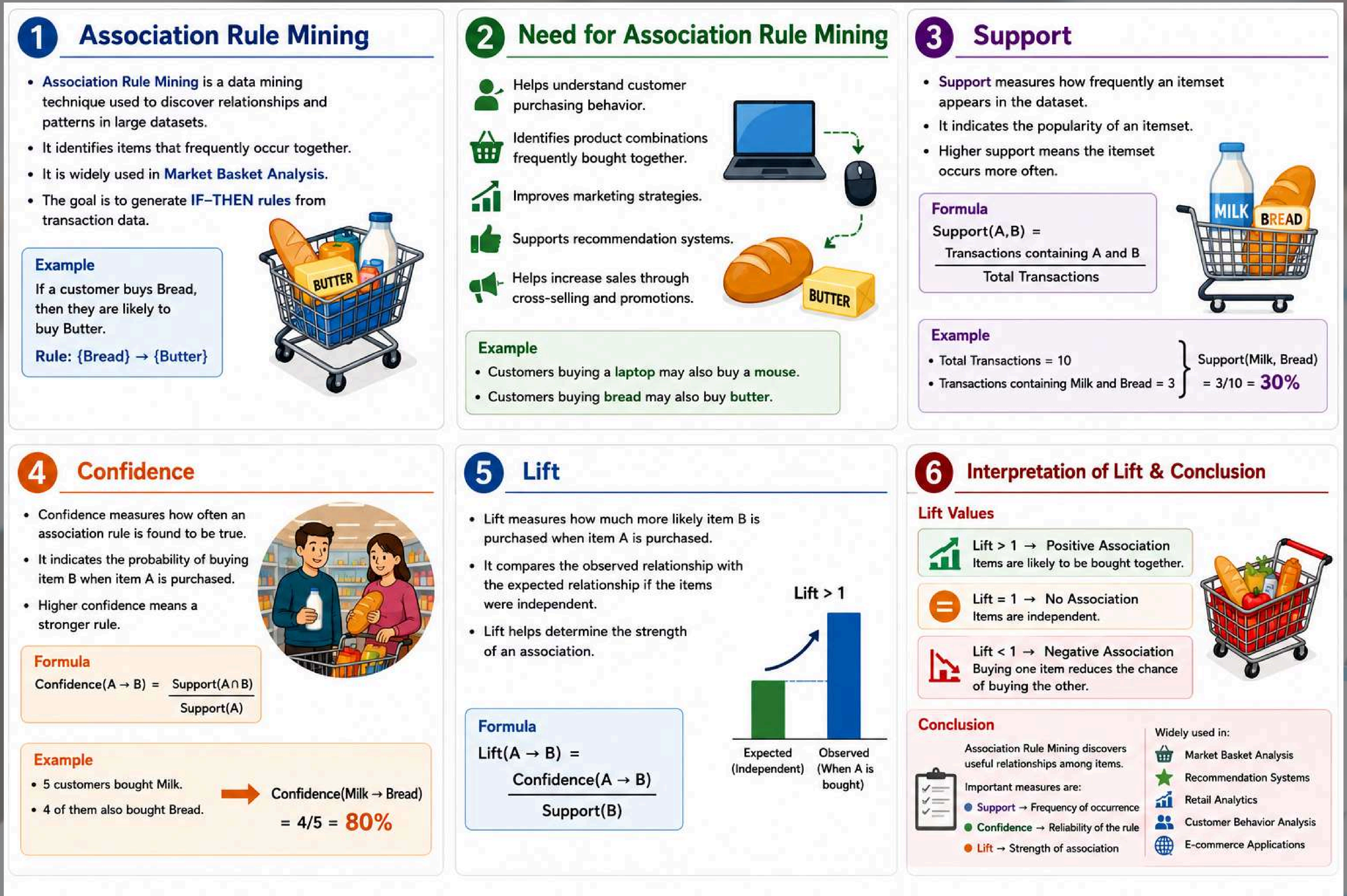
(if they satisfy the minimum support threshold)

Apriori Algorithm

Applications of Apriori Algorithm

1) Market Basket Analysis
   - Identifies products frequently bought together.
   - Helps in store layout planning and promotions.
2) Recommendation Systems
   - Suggests related products to customers.
   - Example: "Customers who bought this item also bought..."
3) Cross-Selling
   - Helps businesses increase sales by recommending complementary products.
4) Customer Behavior Analysis
   - Understands shopping patterns and preferences.

Association Rule Mining



Decision Tree

- A Decision Tree is a Machine Learning algorithm used for both classification and regression problems.
- It works like a flowchart.
- Each internal node represents a decision based on a feature.
- Each branch represents the outcome of that decision.
- Each leaf node represents the final prediction or result.
- Decision Trees are easy to understand and interpret.

Structure of a Decision Tree

Components of a Decision Tree

1. Root Node
   - Starting point of the tree.
   - Represents the first decision.
2. Internal Nodes
   - Represent tests on features.
   - Split data into smaller groups.
3. Branches
   - Show outcomes of decisions.
4. Leaf Nodes
   - Final prediction or classification result.

Decision Tree

Working of Decision Tree

Step 1

- Select the best feature to split the data.

Step 2

- Divide the dataset into smaller groups.

Step 3

- Repeat the splitting process for each group.

Step 4

- Continue until a stopping condition is reached.
- Stopping Conditions
- All records belong to the same class.
- Maximum tree depth is reached.
- No further meaningful split is possible.

Attribute Selection Measures

Decision Trees use measures to choose the best feature for splitting.

1) Information Gain

- Measures the reduction in uncertainty after a split.
- Higher Information Gain indicates a better split.

Decision Tree

2) Gini Index

- Measures impurity in the dataset.
- Lower Gini Index indicates purer groups.

These measures help create an efficient and accurate tree.

Case Study – Loan Approval Prediction

Problem

A bank wants to predict whether a loan should be approved.

Features Used

- Credit Score
- Income
- Employment Status
- Loan Amount

Decision Tree

Target Variable

- Loan Approved (Yes/No)

The Decision Tree learns patterns from previous loan applications and predicts whether a new application should be approved.

Loan Approval Decision Tree Example

Decision Process

```
        [Credit Score > 700?]
             /        \
          Yes          No
           |            |
     [Income?]    [Loan Denied]
            /   \
         High   Low
          |       |
[Loan Approved] [Further Check]
```

Difference between classificationa and clustering

Classification	Clustering
Classification is a supervised learning technique.	Clustering is an unsupervised learning technique.
Uses labeled data for training.	Uses unlabeled data for grouping.
Predicts predefined classes or categories.	Finds natural groups or clusters in data.
Output is a class label (e.g., Pass/Fail, Spam/Not Spam).	Output is a set of clusters of similar data points.
Examples: Decision Tree, Naive Bayes, Logistic Regression.	Examples: K-Means, Hierarchical Clustering, DBSCAN.
Used for prediction and classification tasks.	Used for pattern discovery and data segmentation.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